

The Utility Problem

Asset management focuses on maintenance, rehabilitation, and replacement of infrastructure with the goal of extending the service life and improving the quality of existing infrastructure. It also ensures that the subsequent risk of failure is minimized.

Previously (without data analysis), decisions about asset management activities were mostly based on the age of the asset alone. However, this view is very limited and does not take into account the actual condition of the asset. Some older assets may be in better condition than newer ones because they may be better manufactured, they may be in a different environment, they may be exposed to different influences, etc.

Therefore, by taking a more detailed view of each asset or group of assets, it is possible to extend the life of some assets and thus save significant investment costs for new assets. So this is a very important part of any utility as it is a significant investment and operating cost, but with the proper Asset Management, decisions are made cost-effectively. Also, greater customer satisfaction can be achieved (for example, fewer outages because the asset is in good condition).

Many utilities find it challenging to start addressing Asset Management correctly as it is a very complex issue that requires data from different company systems. However, the complexity of this critical electricity distribution topic should not divert attention.

The Data Solutions and Methods

Asset management should cover all the elements in the grid, but this is very difficult in reality, as it is almost impossible to obtain insightful data about every element. Therefore, the model proposed in this document focuses on larger parts of the grid - in this case, the entire HV/MV or MV/LV substations. Because of this, it is suitable for initial identification of substations and then each substation should be assessed in more detail. During this detailed assessment, it may be found, for example, that only a specific part of a substation with poor metrics needs to be replaced or refurbished and then this substation will regain a good rating.

Several variables (criteria) enter into the calculation:

- 1) Risk Value:
 - a) Technical condition - this includes periodic revisions, checks, measurements and physical inspections of substation components (a single mark is created based on the weights of each criterion).
 - b) Failures (based on real faults in the grid):
 - i) Statistical fault rate - for each type of substation (for example, different substation manufacturers have different failure rates), its probable failure rate is calculated.

- ii) Own fault rate - number of faults according to reported faults for each substation.
- c) Age.
- 2) Importance - what is powered by the substation:
 - a) Number of customers and consumption value.
 - b) Type of customer - for example, hospitals are more critical than households.
- 3) Strategy - the company's strategic decision.
- 4) Operational Expenditures (OPEX) - repair costs.

The Awesense EDM data model contains the parameters mentioned above and it is very easy to get this data. For some of the criteria, some forms of GIS data tracing are used, which are part of the model (e.g. Importance - for this criterion, it is necessary to know the numbers and types of customers that belong to the selected substation and for this the tracing function is needed).

Calculation:

To make the model flexible and adjustable, the resulting values must be within a certain range - Risk Value, Importance, Strategy and OPEX are always in the range 0-1, which is the standard normalization procedure. These four values are multiplied together for the overall Renewability Index. This value is, therefore, again in the 0-1 interval.

$$\text{Renewability Index} < 0; 1 > = \text{Risk Value} \times \text{Importance} \times \text{Strategy} \times \text{OPEX}$$

Renewability Index Formula

$$\text{Risk Value} = (W_{TC} \times \text{Technical condition} + W_{SF} \times \text{Statistical failure} + W_{OF} \times \text{Own failures} + W_A \times \text{Age})/4$$

Risk Value Formula

Weight	Value
W_{TC}	0,25
W_{SF}	0,25
W_{OF}	0,25
W_A	0,25

$$Importance = (W_{NC} \times \text{Number of customers} + W_{IC} \times \text{Importance of customers})/4$$

Importance Formula

Weight	Value
W_{NC}	0,25
W_{IC}	0,25

$$Strategy = \text{Strategic decision} / 4$$

Strategy Formula

$$OPEX = \text{Costs} / 4$$

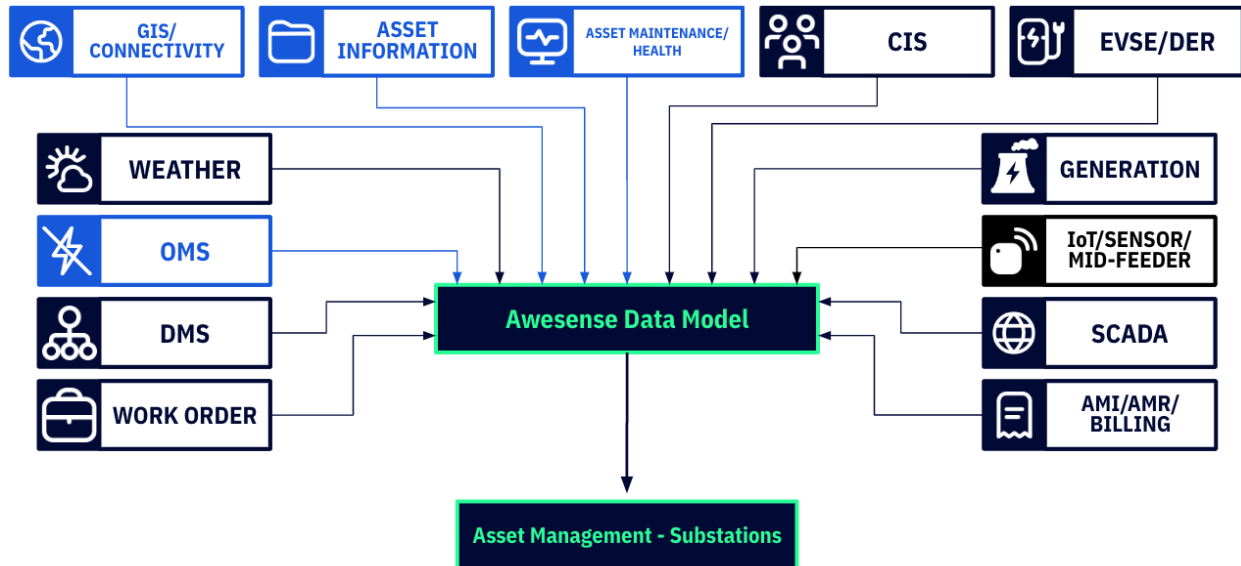
Opex Formula

Technical condition, Statistical failure, Own failures, Age, Number of customers and Importance of customers, Strategic decision and Costs have marks in the interval 1-4.

Historical data is taken from recent years or decades (depending on the criterion). The more data available, the better. If some data or variables are unavailable, a neutral value is used, which will not affect the calculation (will not be taken into account in the calculation). This can be used if the data is challenging to get. Any parameter that meets the essential criteria specified by the Awesense calculation methodology can be added to the calculation. All this data can be assigned to each substation. The final results are plotted on a map (using a heat map) for better visualization and for easier identification of problem locations.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in royal blue are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.